Time Series: Autoregressive models AR, MA, ARMA, ARIMA

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ARIMA Models

- ARIMA is an acronym that stands for **A**uto-**R**egressive **I**ntegrated **M**oving **A**verage. Specifically,
  - **AR** *Autoregression.* A model that uses the dependent relationship between an observation and some number of lagged observations.
  - **I** *Integrated.* The use of differencing of raw observations in order to make the time series stationary.
  - **MA** *Moving Average.* A model that uses the dependency between an observation and a residual error from a moving average model applied to lagged observations.
- Each of these components are explicitly specified in the model as a parameter.
- Note that **AR** and **MA** are two widely used linear models that work on stationary time series, and **I** is a preprocessing procedure to "stationarize" time series if needed.

Notations

- A standard notation is used of $ARIMA(p, d, q)$ where the parameters are substituted with integer values to quickly indicate the specific ARIMA model being used.
  - **p** The number of lag observations included in the model, also called the lag order.
  - **d** The number of times that the raw observations are differenced, also called the degree of differencing.
  - **q** The size of the moving average window, also called the order of moving average.
- A value of 0 can be used for a parameter, which indicates to not use that element of the model.
- In other words, ARIMA model can be configured to perform the function of an ARMA model, and even a simple AR, I, or MA model.

Autoregressive Models

- **Intuition**
  - Autoregressive models are based on the idea that current value of the series, X_t, can be explained as a linear combination of p past values, X_{t-1}, X_{t-2}, ..., X_{t-p}, together with a random error in the same series.
- **Definition**
  - An autoregressive model of order p, abbreviated $AR(p)$, is of the form

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + w_t = \sum_{i=1}^{p} \phi_i X_{t-i} + w_t$$

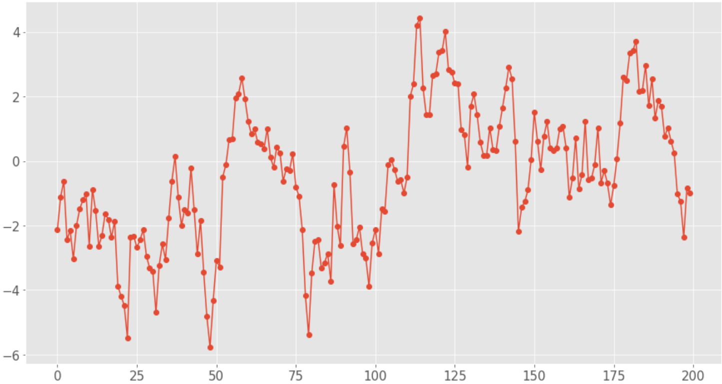
    where X_t is stationary, $w_t \sim wn(0, \sigma_w^2)$, and ϕ_1, ϕ_2, ..., ϕ_p ($\phi_p \neq 0$) are model parameters. The hyperparameter p represents the length of the "direct look back" in the series.

AR Example: AR(0) and AR(1)

- The simplest AR process is $AR(0)$, which has no dependence between the terms. In fact, $AR(0)$ is essentially white noise.
- $AR(1)$ can be given by $X_t = \phi_1 X_{t-1} + w_t$.
  - Only the previous term in the process and the noise term contribute to the output.
  - If $|\phi_1|$ is close to 0, then the process still looks like white noise.
  - If $\phi_1 < 0$, X_t tends to oscillate between positive and negative values.
  - If $\phi_1 = 1$ then the process is equivalent to random walk, which is not stationary as the variance is dependent on t (and infinite).

AR Examples: AR(1) Process

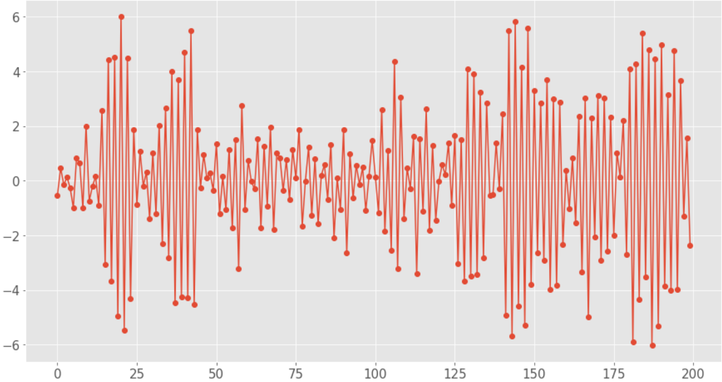
- Simulated $AR(1)$ Process $X_t = 0.9X_{t-1} + w_t$:



- **Mean** $E[X_t] = 0$
- **Variance** $\text{Var}(X_t) = \dfrac{\sigma_w^2}{(1 - \phi_1^2)}$

AR Examples: AR(1) Process

- Simulated $AR(1)$ Process $X_t = -0.9X_{t-1} + w_t$:



- **Mean** $E[X_t] = 0$
- **Variance** $\mathrm{Var}(X_t) = \dfrac{\sigma_w^2}{(1-\phi_1^2)}$

Moving Average Models (MA)

- The name might be misleading, but moving average models should not be confused with the moving average smoothing.
- **Motivation**
  - Recall that in AR models, current observation X_t is regressed using the previous observations $X_{t-1}, X_{t-2}, \ldots, X_{t-p}$, plus an error term w_t at current time point.
  - One problem of AR model is the ignorance of correlated noise structures (which is unobservable) in the time series.
  - In other words, the imperfectly predictable terms in current time, w_t, and previous steps, $w_{t-1}, w_{t-2}, \ldots, w_{t-q}$, are also informative for predicting observations.

Moving Average Models (MA)

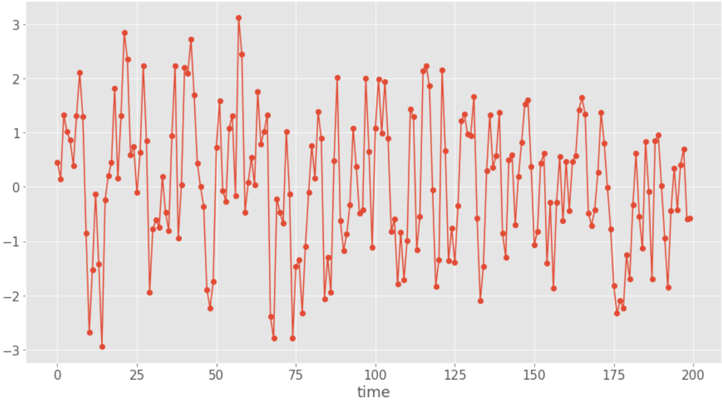
- **Definition**
  - A moving average model of order q, or $MA(q)$, is defined to be

$$X_t = w_t + \theta_1 w_{t-1} + \theta_2 w_{t-2} + \cdots + \theta_q w_{t-q} = w_t + \sum_{j=1}^{q} \theta_j w_{t-j}$$

    where $w_t \sim wn(0, \sigma_w^2)$, and $\theta_1, \theta_2, \ldots, \theta_q$ $(\theta_q \neq 0)$ are parameters.
  - Although it looks like a regression model, the difference is that the w_t is not observable.
- Contrary to AR model, finite MA model is always stationary, because the observation is just a weighted moving average over past forecast errors.

MA Examples: MA(1) Process

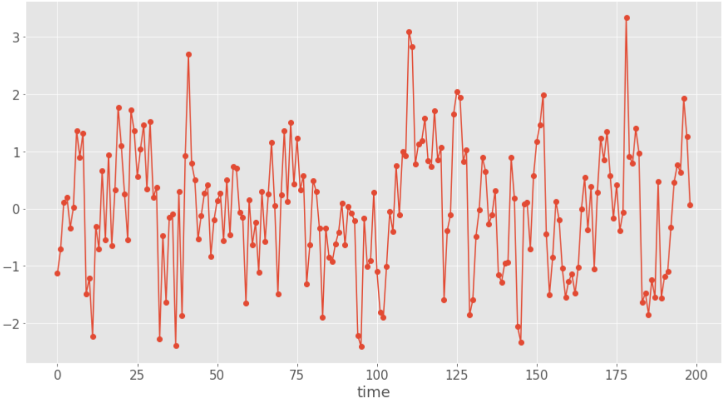
- Simulated $MA(1)$ Process $X_t = w_t + 0.8 \times w_{t-1}$:



- **Mean** $E[X_t] = 0$
- **Variance** $\text{Var}(X_t) = \sigma_w^2(1 + \theta_1^2)$

MA Examples: MA(2) Process

- Simulated $MA(2)$ Process $X_t = w_t + 0.5 \times w_{t-1} + 0.3 \times w_{t-2}$:



- **Mean** $E[X_t] = 0$
- **Variance** $\mathrm{Var}(X_t) = \sigma_w^2(1 + \theta_1^2 + \theta_2^2)$

ARMA Models

- Autoregressive and moving average models can be combined together to form ARMA models.
- **Definition**
  - A time series $\{x_t; t = 0, \pm 1, \pm 2, \dots\}$ is $ARMA(p, q)$ if it is stationary and

$$X_t = w_t + \sum_{i=1}^{p} \phi_i X_{t-i} + \sum_{j=1}^{q} \theta_j w_{t-j},$$

    where $\phi_p \neq 0$, $\theta_q \neq 0$, and $\sigma_w^2 > 0$, $w_t \sim wn(0, \sigma_w^2)$.
  - With the help of AR operator and MA operator we defined before, the model can be rewritten more concisely as:

$$\boxed{\phi(B)X_t = \theta(B)w_t}$$

"Stationarize" Nonstationary Time Series

- One limitation of ARMA models is the stationarity condition.
- In many situations, time series can be thought of as being composed of two components, a non-stationary trend series and a zero-mean stationary series, i.e. $X_t = \mu_t + Y_t$.
- **Strategies**
  - Detrending: Subtracting with an estimate for trend and deal with residuals.

$$\hat{Y}_t = X_t - \hat{\mu}_t$$

  - Differencing: Recall that random walk with drift is capable of representing trend, thus we can model trend as a stochastic component as well.

$$\mu_t = \delta + \mu_{t-1} + w_t$$
$$\nabla X_t = X_t - X_{t-1} = \delta + w_t + (Y_t - Y_{t-1}) = \delta + w_t + \nabla Y_t$$

    ∇ is defined as the first difference and it can be extended to higher orders.

From ARMA to ARIMA

- **Order of Differencing**
  - Differences of order d are defined as

$$\nabla^d = (1 - B)^d,$$

where $(1 - B)^d$ can be expanded algebraically for higher integer values of d.

- **Definition**
  - A process X_t is said to be $ARIMA(p, d, q)$ if

$$\nabla^d X_t = (1 - B)^d X_t$$

is $ARMA(p, q)$.

  - In general, $ARIMA(p, d, q)$ model can be written as:

$$\phi(B)(1 - B)^d X_t = \theta(B) w_t$$

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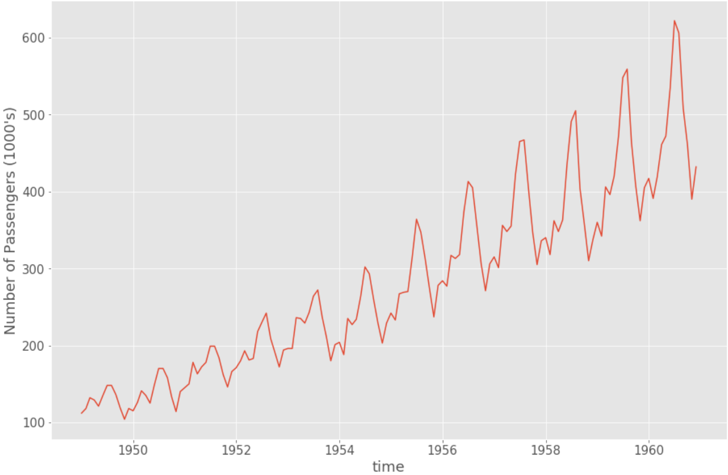
Box-Jenkins Methods

- As we have seen ARIMA models have numerous parameters and hyper parameters, Box and Jenkins suggests an iterative three-stage approach to estimate an ARIMA model.
- **Procedures**
  1. Model identification: Checking stationarity and seasonality, performing differencing if necessary, choosing model specification $ARIMA(p, d, q)$.
  2. Parameter estimation: Computing coefficients that best fit the selected ARIMA model using *maximum likelihood estimation* or *non-linear least-squares estimation*.
  3. Model checking: Testing whether the obtained model conforms to the specifications of a stationary univariate process (i.e. the residuals should be independent of each other and have constant mean and variance). If failed go back to step 1.
- Let's go through a concrete example together for this procedure.

Air Passenger Data



Monthly totals of a US airline passengers, from 1949 to 1960

Stationarity Test: ADF Test

- Results of Augmented Dickey-Fuller Test

Item	Value
Test Statistic	0.815369
p-value	0.991880
#Lags Used	13.000000
Number of Observations Used	130.000000
Critical Value (1%)	-3.481682
Critical Value (5%)	-2.884042
Critical Value (10%)	-2.578770

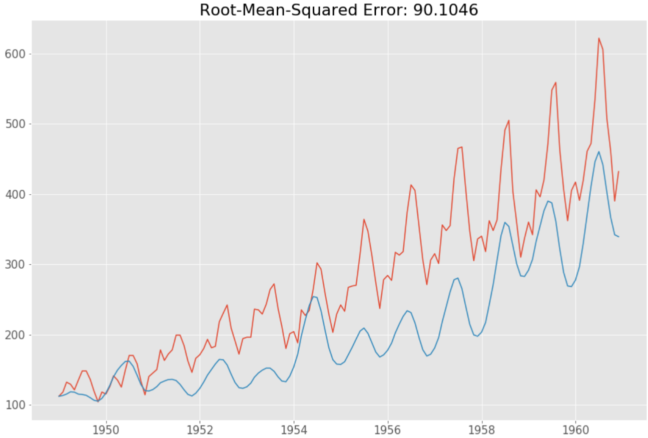
- The test statistic is a negative number.
- The more negative it is, the stronger the rejection of the null hypothesis.

Stationarize Time Series

- As all previous methods show that the initial time series is non-stationary, it's necessary to perform transformations to make it stationary for ARMA modeling.
  - Detrending
  - Differencing
  - Transformation: Applying arithmetic operations like log, square root, cube root, etc. to stationarize a time series.
  - Aggregation: Taking average over a longer time period, like weekly/monthly.
  - Smoothing: Removing rolling average from original time series.
  - Decomposition: Modeling trend and seasonality explicitly and removing them from the time series.

Forecasting

- The last step is to reverse the transformations we've done to get the prediction on original scale.



SARIMA: Seasonal ARIMA Models

- One problem in the previous model is the lack of seasonality, which can be addressed in a generalized version of ARIMA model called seasonal ARIMA.
- **Definition**
  - A seasonal ARIMA model is formed by including additional seasonal terms in the ARIMA models, denoted as

    $$ARIMA\,(p, d, q)\,(P, D, Q)_m,$$

    i.e.

    $$\phi(B^m)\,\phi(B)\,(1 - B^m)^D\,(1 - B)^d\,X_t = \theta(B^m)\,\theta(B)\,w_t$$

    where m represents the number of observations per year.
  - The seasonal part of the model consists of terms that are similar to the non-seasonal components, but involve backshifts of the seasonal period.